

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Pravin R (192421292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability. (5 Marks)

6. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

7. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

8. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

9. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

10. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection,

crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Functional Blocks of a Machine Learning System

A Machine Learning (ML) system consists of the following functional blocks:

1. Data Collection

- Collect data from sensors, databases, websites, or files.
- Example: Collecting patient health records.

2. Data Preprocessing

- Remove missing values, duplicate records, and noise.
- Normalize and encode categorical data.

3. Feature Extraction

- Select important attributes that influence prediction.
- Example: Age, blood pressure, glucose level.

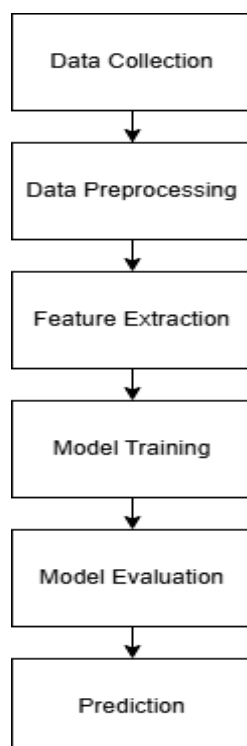
4. Model Training

- Train the algorithm using training data.
- Supervised learning uses labelled data.
- Unsupervised learning uses unlabelled data.

5. Model Evaluation

- Test the trained model using unseen data.
- Metrics include Accuracy, Precision, Recall, and F1-score.

Working:



Examples

- **Supervised Learning:** Email spam detection using labelled emails.
- **Unsupervised Learning:** Customer segmentation using purchasing behaviour without labels.

2. Machine Learning System for Disaster Management

- A machine learning system for disaster management predicts floods, earthquakes, and other natural disasters using historical and real-time data.
- The input data includes rainfall, river water levels, weather conditions, soil moisture, and seismic sensor readings. A Long Short-Term Memory (LSTM) Neural Network is suitable because it can learn patterns from time-series data. Random Forest can also be used for classification tasks.
- Before training, data preprocessing is performed by removing missing values, eliminating duplicate records, and normalizing the data. Important features such as rainfall, river level, and seismic activity are selected to improve prediction accuracy.
- The dataset is divided into training, validation, and testing sets. The model is evaluated using Accuracy, Precision, Recall, and F1-score. The system provides early warnings, supports risk assessment, and helps authorities plan evacuation and emergency response efficiently.

Applications

- Early warning systems
- Flood prediction
- Earthquake risk estimation
- Emergency evacuation planning
- Resource allocation

3. Backpropagation in Neural Networks

Backpropagation is a supervised learning algorithm used to train neural networks by updating the weights to minimize prediction errors.

During the forward pass, input values are multiplied by weights and passed through activation functions to produce an output. The predicted output is then compared with the target output, and the error is calculated.

In the backward pass, this error is propagated backward through the network. The weights are updated using the formula:

$$\text{New Weight} = \text{Old Weight} + \text{Learning Rate} \times \text{Error} \times \text{Input}$$

The learning rate determines the size of weight updates. A very small learning rate slows training, while a very large value may prevent convergence. Activation functions such as ReLU, Sigmoid, and Tanh introduce non-linearity, allowing the neural network to learn complex relationships and improve classification accuracy.

4. Genetic Algorithm Evolution

A Genetic Algorithm (GA) is an optimization technique inspired by natural evolution. It improves candidate solutions through several generations.

Initially, the population consists of candidate solutions with different fitness values. During selection, individuals with the highest fitness values are chosen as parents. From the given dataset, solutions with fitness values 80 and 75 are selected.

During crossover, the selected parents exchange their characteristics to generate new offspring with improved fitness. Next, mutation randomly changes one or more genes to introduce diversity and avoid local optimum solutions.

The population evolves over multiple generations, producing increasingly better solutions. Finally, the chromosome with the highest fitness value is selected as the optimal solution. In this dataset, the best solution is High Speed and High Accuracy, with the maximum fitness after evolution.

5. Neural Network Classification

The given dataset is used to train a neural network that predicts whether a patient has a disease based on Age, Blood Pressure, Glucose Level, and Lifestyle Risk.

From the dataset:

- **True Positive (TP) = 2**
- **True Negative (TN) = 1**
- **False Positive (FP) = 1**
- **False Negative (FN) = 1**

The evaluation metrics are:

- **Accuracy** = $(TP + TN) / \text{Total} = (2 + 1) / 5 = 60\%$
- **Precision** = $TP / (TP + FP) = 2 / 3 = 66.67\%$
- **Recall** = $TP / (TP + FN) = 2 / 3 = 66.67\%$
- **F1-score** = 66.67%

Data preprocessing removes missing values and normalizes numerical data. Feature selection identifies important attributes such as Glucose Level and Blood Pressure, reducing unnecessary features and improving model accuracy and reliability.

6. Machine Learning System for Smart Traffic Management

A smart traffic management system predicts traffic congestion using machine learning.

The input features include vehicle density, weather conditions, road type, traffic signals, and time of day. Random Forest is a suitable algorithm because it handles large datasets, mixed data types, and provides high prediction accuracy.

Before training, preprocessing removes missing values, eliminates duplicates, normalizes numerical values, and encodes categorical variables. Feature engineering creates additional features such as rush-hour indicators and weekend flags.

The model is trained using historical traffic data and validated using testing data. Performance is measured using Accuracy, Precision, Recall, and F1-score. After deployment, the system predicts congestion in real time, optimizes traffic signals, suggests alternate routes, and improves emergency vehicle movement.

7. Convolutional Neural Networks (CNNs)

A Convolutional Neural Network (CNN) is a deep learning model mainly used for image classification and object recognition.

The Convolution Layer applies filters to the input image and extracts features such as edges, textures, and shapes. The Pooling Layer reduces the size of feature maps, decreasing computation while retaining important information. The Fully Connected Layer combines extracted features and performs final classification.

During training, backpropagation calculates prediction errors and updates the filter weights using gradient descent. For example, when classifying an image of a cat, the CNN extracts important features and predicts the correct class.

The performance of a CNN depends on factors such as dataset size, learning rate, number of filters, batch size, number of epochs, and image quality.

8. Genetic Algorithm for Feature Selection

Genetic Algorithms can be used for selecting the most relevant features in a machine learning model.

Each feature subset is represented as a binary chromosome, where 1 indicates the feature is selected and 0 indicates it is not selected. The fitness function evaluates each chromosome based on classification accuracy and the number of selected features.

The selection process chooses chromosomes with higher fitness values. Crossover combines two parent chromosomes to produce improved offspring, while mutation randomly changes bits to introduce diversity and avoid premature convergence.

After several generations, the algorithm identifies the best feature subset. Feature selection reduces model complexity, decreases training time, prevents overfitting, and improves prediction accuracy by removing irrelevant features.

9. Evaluation Metrics for Classification

Classification models are evaluated using a confusion matrix containing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

The commonly used evaluation metrics are:

- **Accuracy:** Percentage of correctly classified instances.
- **Precision:** Percentage of correct positive predictions.
- **Recall:** Percentage of actual positives correctly identified.
- **F1-score:** Harmonic mean of Precision and Recall.
- **ROC Curve:** Graph showing the relationship between True Positive Rate and False Positive Rate. A larger Area Under Curve (AUC) indicates better performance.

These metrics are especially important in applications such as medical diagnosis, where false negatives are critical, and fraud detection, where minimizing false positives improves system reliability.

10. Genetic Algorithm for Scheduling Optimization

Genetic Algorithms are widely used to solve scheduling problems by finding near-optimal task sequences.

Each schedule is represented as a chromosome, where each gene represents the order of a task. The fitness function evaluates the quality of a schedule based on execution time, waiting time, resource utilization, and overall completion time.

During selection, the best schedules are chosen as parents. Crossover combines parent schedules to generate better offspring, while mutation randomly changes task positions to explore new solutions and avoid local optima.

After several generations, the algorithm produces an optimized schedule with improved efficiency. Genetic Algorithms are effective for solving complex scheduling problems because they search large solution spaces quickly and provide high-quality solutions with reasonable computational time.